13. The Jackknife 2

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Recall the Motivation: We would like to replicate a data set multiple times.

Jackknife: Create n Leave-One-Out datasets

$$\left.\begin{array}{l} (X_2, \dots, X_n) \to \hat{\theta}_1 \\ (X_1, X_3, \dots, X_n) \to \hat{\theta}_2 \\ \vdots \\ (X_1, X_2, \dots, X_{n-1}) \to \hat{\theta}_n \end{array}\right\} \quad \hat{\theta}_{Jack} = \frac{1}{n}\sum_{i=1}^{n} \hat{\theta}_i$$

Average of the LOO estimates

Original Estimate

$$(X_1, \dots, X_n) \to \hat{\theta}$$

Difference between Original and Jackknife estimate

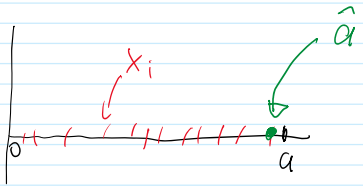
Bias: $\hat{b}(\hat{\theta})_{Jack} = (n-1)(\hat{\theta}_{Jack} - \hat{\theta})$

De-biased Est: $\hat{\theta} - \hat{b}(\hat{\theta})_{Jack} = n\hat{\theta} - (n-1)\hat{\theta}_{Jack}$

Example: Uniform$[0, \overset{\theta}{a}]$ with a unknown

* MLE: $\hat{a} = \max_{i=1,\dots,n} \{X_i\}$

always biased as $\hat{a} < a$ (a.s.)



Order Statistics:

- if we have $X_1, \dots, X_n$ real (cont) r.v.s then $X_{(1)} < X_{(2)} < \dots < X_{(n)}$ are the order statistics.

  $\therefore \hat{a} = X_{(n)}$

  $\mathbb{E} X_{(k)} = \frac{k}{k + (n-k+1)}$

- For $X_i \sim \text{Unif}[0,1]$ then $X_{(k)} \sim \text{Beta}(k, n-k+1)$

- For simplicity, assume $a = 1$.

$$\to \text{Bias}(\hat{a}) = \underbrace{\mathbb{E} X_{(n)}}_{\mathbb{E}\hat{\theta}} - \underbrace{1}_{\theta} = \frac{n}{n+1} - 1 = \boxed{\frac{-1}{n+1}} = O(n^{-1})$$

Bias of the MLE $\hat{a}$ for $a = 1$

$$\hat{a}_{Jack} = \frac{1}{n}\sum_{i=1}^{n} \underbrace{\hat{a}_i}_{\text{LOO}}, \quad \hat{a}_i = \begin{cases} X_{(n)}, & \text{if we didn't remove } X_{(n)} \\ X_{(n-1)}, & \text{if we did remove } X_{(n)} \end{cases}$$

$$\hat{a}_{Jack} = \frac{1}{n}\sum_{i=1} \underset{\text{LOO}}{a_i}, \quad a_i = \begin{cases} \\ X_{(n-1)}, & \text{if we did remove } X_{(n)} \end{cases}$$

$$= \frac{1}{n}\Big[\underbrace{X_{(n)} + \dots + X_{(n)}}_{n-1} + X_{(n-1)}\Big]$$

if I remove $X_{(n)}$

$$= \left(\frac{n-1}{n}\right) X_{(n)} + \frac{1}{n} X_{(n-1)}$$

max element

2nd largest element

Note: $\hat{a}_{Jack} < \hat{a} < a$ (a.s.)

Debias: $n\hat{a} - (n-1)\hat{a}_{Jack}$

$$= n X_{(n)} - (n-1)\left[\left(\frac{n-1}{n}\right) X_{(n)} + \frac{1}{n} X_{(n-1)}\right]$$

$$= \frac{n^2}{n} X_{(n)} - \left(\frac{n^2-2n+1}{n}\right) X_{(n)} - \left(\frac{n-1}{n}\right) X_{(n-1)}$$

$$= \left(\frac{2n-1}{n}\right) X_{(n)} - \left(\frac{n-1}{n}\right) X_{(n-1)} = \underset{\hat{a}}{2X_{(n)}} - \left(\frac{1}{n} X_{(n)} + \left(\frac{n-1}{n}\right) X_{(n-1)}\right)$$

$$\text{Bias}(\hat{a}) = \boxed{\frac{-1}{n+1}} = O(n^{-1})$$

$$\text{Bias}(\text{Debiased Est}) = \mathbb{E}\left[2X_{(n)} - \frac{1}{n} X_{(n)} - \left(\frac{n-1}{n}\right) X_{(n-1)}\right] - \overset{a}{1}$$

$$= 2\left(\frac{n}{n+1}\right) - \frac{1}{n}\left(\frac{n}{n+1}\right) - \left(\frac{n-1}{n}\right)\left(\frac{n-1}{n+1}\right) - 1$$

$$= \frac{1}{n+1}\left[2n - 1 - \frac{n^2-2n+1}{n} - n - 1\right]$$

$$= \frac{1}{n(n+1)}\left[\cancel{n^2} - \cancel{2n} - \cancel{n^2} + \cancel{2n} - 1\right] = \boxed{\frac{-1}{n(n+1)}} = O(n^{-2})$$

$\mathbb{E}X_{(n)} = \frac{n}{n+1}$

$\mathbb{E}X_{(n-1)} = \frac{n-1}{n+1}$

$\vdots$

$\mathbb{E}X_{(1)} = \frac{1}{n+1}$

Now, the bias decreases at a faster rate!

But what about the Variance and MSE?